PG No. 39 $S_4(c)$ $\bar{4}$ [tetragonal]

Table 1: Representation matrices

Irrep.				
A	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^{+}_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^{-}_{001}:\begin{bmatrix}1\end{bmatrix}$
B	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^{+}_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^{-}_{001}:\begin{bmatrix}-1\end{bmatrix}$
$E^{(a)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^{+}_{001}:\begin{bmatrix}i\end{bmatrix}$	$-4^{-}_{001}:\begin{bmatrix}-i\end{bmatrix}$
$E^{(b)}$	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^{+}_{001}:\begin{bmatrix}-i\end{bmatrix}$	$-4^{-}_{001}:\begin{bmatrix}i\end{bmatrix}$